

## WUZHOU, China

WMO: 592650

Lat: 23.4783N

Lon: 111.3037E

Elev: 120

StdP: 99.89

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 3.9        | 5.2 | -3.0                          | 3.0 | 10.6 | -0.7 | 3.6 | 9.1  | 5.4                   | 9.8  | 5.0 | 9.6  | 2.1                   | 40   | 0.302 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.3                    | 34.8            | 26.6 | 34.1 | 26.4 | 33.3 | 26.3 | 27.8                | 32.4 | 27.4 | 31.9 | 27.1 | 31.3 | 2.0                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.7                            | 22.6 | 29.8 | 26.3 | 22.1 | 29.4 | 26.0 | 21.7 | 29.2 | 90.0          | 32.6 | 88.2 | 32.1 | 86.5 | 31.5 | 29.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 5.0               | 4.2  | 3.5 | 2.1                        | 36.4 | 1.5                | 0.6 | 1.1                                                | 36.8 | 0.2        | 37.2 | -0.6       | 37.6 | -1.7       | 38.0 |
|                   |      |     | 0.7                        | 28.6 | 1.2                | 0.6 | -0.2                                               | 29.0 | -0.9       | 29.3 | -1.6       | 29.6 | -2.5       | 30.0 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 22.0   | 12.7  | 14.6  | 17.5  | 22.4  | 25.9  | 27.9  | 28.9  | 28.7  | 27.3  | 24.1  | 19.4  | 14.3  |
|                                                                        | DBStd       | 6.67   | 4.47  | 5.18  | 4.46  | 3.77  | 2.48  | 1.83  | 1.60  | 1.58  | 2.10  | 2.65  | 3.53  | 3.63  |
|                                                                        | HDD10.0     | 44     | 22    | 14    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 6     |
|                                                                        | HDD18.3     | 557    | 183   | 126   | 72    | 12    | 0     | 0     | 0     | 0     | 0     | 1     | 29    | 134   |
|                                                                        | CDD10.0     | 4426   | 105   | 142   | 235   | 372   | 492   | 537   | 585   | 581   | 518   | 438   | 281   | 139   |
|                                                                        | CDD18.3     | 1896   | 7     | 21    | 46    | 133   | 234   | 287   | 327   | 323   | 268   | 181   | 60    | 9     |
|                                                                        | CDH23.3     | 17752  | 19    | 105   | 211   | 872   | 1976  | 2933  | 3684  | 3577  | 2672  | 1343  | 328   | 33    |
| (13)                                                                   | CDH26.7     | 6718   | 1     | 15    | 31    | 233   | 681   | 1141  | 1589  | 1518  | 1040  | 413   | 56    | 1     |
| (14)                                                                   | Wind        | WSAvg  | 1.7   | 1.8   | 1.8   | 1.8   | 1.7   | 1.6   | 1.7   | 1.6   | 1.7   | 1.8   | 1.7   | 1.8   |
| (15) Precipitation                                                     | PrecAvg     | 1487   | 52    | 69    | 81    | 222   | 262   | 208   | 166   | 193   | 98    | 67    | 41    | 27    |
|                                                                        | PrecMax     | 1907   | 224   | 270   | 220   | 517   | 407   | 709   | 377   | 453   | 275   | 226   | 175   | 91    |
|                                                                        | PrecMin     | 1035   | 2     | 3     | 11    | 102   | 32    | 54    | 48    | 59    | 11    | 3     | 0     | 1     |
|                                                                        | PrecStd     | 223    | 52    | 67    | 47    | 99    | 90    | 139   | 86    | 102   | 67    | 50    | 43    | 25    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 25.4  | 28.7  | 29.5  | 32.5  | 34.2  | 35.0  | 35.8  | 35.6  | 34.6  | 32.8  | 30.0  | 26.0  |
|                                                                        |             | MCWB   | 19.2  | 22.1  | 22.9  | 25.1  | 26.1  | 26.8  | 26.9  | 26.8  | 26.4  | 24.4  | 22.9  | 20.2  |
|                                                                        | 2%          | DB     | 23.4  | 26.2  | 27.3  | 30.7  | 32.7  | 34.0  | 34.9  | 34.6  | 33.6  | 31.4  | 28.2  | 24.2  |
|                                                                        |             | MCWB   | 18.3  | 20.2  | 22.1  | 24.4  | 25.7  | 26.6  | 26.6  | 26.6  | 26.0  | 23.7  | 21.3  | 18.8  |
|                                                                        | 5%          | DB     | 21.5  | 24.4  | 25.7  | 29.3  | 31.6  | 33.0  | 33.9  | 33.8  | 32.7  | 30.3  | 26.9  | 22.1  |
|                                                                        |             | MCWB   | 16.8  | 19.3  | 21.0  | 23.8  | 25.2  | 26.4  | 26.5  | 26.4  | 25.7  | 23.0  | 20.3  | 16.9  |
|                                                                        | 10%         | DB     | 19.6  | 22.5  | 24.1  | 27.9  | 30.4  | 32.0  | 32.9  | 32.8  | 31.6  | 29.1  | 25.3  | 20.3  |
|                                                                        |             | MCWB   | 15.7  | 18.4  | 20.2  | 23.3  | 24.8  | 26.3  | 26.5  | 26.3  | 25.3  | 22.4  | 19.5  | 15.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 20.2  | 22.6  | 24.1  | 26.1  | 27.7  | 28.2  | 28.2  | 28.3  | 27.7  | 26.2  | 24.1  | 22.1  |
|                                                                        |             | MCDB   | 23.8  | 27.4  | 27.9  | 30.6  | 31.9  | 32.6  | 33.0  | 32.9  | 32.0  | 30.5  | 28.1  | 24.6  |
|                                                                        | 2%          | WB     | 19.0  | 21.2  | 22.9  | 25.2  | 26.8  | 27.6  | 27.7  | 27.7  | 27.1  | 25.1  | 23.2  | 20.0  |
|                                                                        |             | MCDB   | 22.4  | 25.1  | 26.3  | 29.3  | 30.9  | 32.0  | 32.4  | 32.3  | 31.4  | 29.2  | 26.5  | 22.7  |
|                                                                        | 5%          | WB     | 17.7  | 20.0  | 21.9  | 24.5  | 26.2  | 27.1  | 27.3  | 27.2  | 26.6  | 24.5  | 22.2  | 18.2  |
|                                                                        |             | MCDB   | 20.7  | 23.4  | 24.6  | 28.2  | 30.0  | 31.4  | 31.7  | 31.6  | 30.6  | 28.2  | 25.4  | 20.9  |
|                                                                        | 10%         | WB     | 16.3  | 18.8  | 20.8  | 23.9  | 25.6  | 26.7  | 26.9  | 26.9  | 26.1  | 23.8  | 21.0  | 16.5  |
|                                                                        |             | MCDB   | 19.2  | 22.0  | 23.4  | 27.1  | 29.1  | 30.6  | 31.1  | 31.0  | 29.8  | 27.2  | 23.8  | 19.7  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.9   | 6.8   | 6.4   | 6.6   | 7.2   | 6.9   | 7.3   | 7.5   | 7.5   | 8.1   | 8.3   | 8.0   |
|                                                                        |             | MCDBR  | 9.8   | 10.0  | 9.0   | 8.8   | 9.1   | 8.6   | 8.8   | 8.9   | 8.8   | 9.4   | 10.0  | 10.2  |
|                                                                        |             | MCWBR  | 5.2   | 4.7   | 4.4   | 3.9   | 3.2   | 2.6   | 2.3   | 2.5   | 2.6   | 3.1   | 4.2   | 5.3   |
|                                                                        | 5% WB       | MCDBR  | 8.4   | 9.0   | 7.9   | 7.8   | 7.8   | 7.6   | 8.0   | 8.0   | 7.8   | 7.7   | 8.3   | 8.0   |
|                                                                        |             | MCWBR  | 4.8   | 4.9   | 4.9   | 4.1   | 3.5   | 3.0   | 2.9   | 2.9   | 2.8   | 3.2   | 4.4   | 5.0   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.635 | 0.687 | 0.831 | 0.799 | 0.625 | 0.578 | 0.552 | 0.623 | 0.678 | 0.721 | 0.606 | 0.611 |
|                                                                        | taud        |        | 1.639 | 1.586 | 1.377 | 1.469 | 1.827 | 1.961 | 2.024 | 1.819 | 1.677 | 1.549 | 1.746 | 1.706 |
|                                                                        | Ebn at Noon |        | 623   | 628   | 566   | 601   | 714   | 742   | 762   | 710   | 658   | 594   | 644   | 622   |
|                                                                        | Edh at Noon |        | 231   | 259   | 333   | 308   | 215   | 186   | 175   | 214   | 243   | 265   | 205   | 209   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.35  | 2.48  | 2.61  | 3.20  | 3.92  | 4.23  | 4.96  | 4.85  | 4.48  | 4.06  | 3.21  | 2.73  |
|                                                                        | RadStd      |        | 0.52  | 0.42  | 0.38  | 0.47  | 0.40  | 0.59  | 0.46  | 0.33  | 0.32  | 0.35  | 0.47  | 0.54  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (6 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +61     |
|                        | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page